

STEVEN DALTON BARNETT

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RESEARCH INTERESTS	Bayesian statistics, computer experiments, surrogate modeling, computer model calibration, Gaussian process regression, and statistical computing. Application areas include engineering, space science, and environmental issues.	
EDUCATION	VIRGINIA POLYTECHNIC INSTITUTE AND STATE UNIVERSITY Ph.D. Statistics, May 2026, advised by Robert B. Gramacy Dissertation: <i>Scalable Surrogates for Counts and Computer Experiments</i> M.S. Statistics, May 2022 BRIGHAM YOUNG UNIVERSITY B.S. Computer Science, April 2016	
PROFESSIONAL POSITIONS	LOS ALAMOS NATIONAL LABORATORY Staff Scientist , March 2026 - Present	
	QUALTRICS, INC Software Engineer II , March 2020 - July 2020 COVID ENGINEERING - Designed and implemented system to track COVID-19 viral tests and notify patients of results - Integrated notification framework with labs throughout the United States to enable batch testing Software Engineer I , July 2016 - March 2020 CLIENT DATA AUTOMATIONS PLATFORM - Designed and built framework for users to schedule repeated automations of bulk data exchange - Developed feature to generate survey response exports and deliver them to customer data stores - Built automations integrated with Amazon Simple Workflow Service, DynamoDB, and Nomad ENGINEERING SERVICES - Built real-time dashboards for managers to complete 360 surveys and view results - Automated the distribution of over 100,000 daily survey invitations to customer contact lists - Scoped, designed, and implemented custom solutions for several Fortune 500 companies	
PEER-REVIEWED PAPERS	O’Leary, E. J., Stringham, D. Barnett, S. D. , & Flowers, A. R. (2026). A Longitudinal Investigation of Concert Band Music Assessment. <i>(To appear in Visions of Research in Music Education)</i> Barnett, S. D. , Beesley, L. J., Booth, A. S., Gramacy, R. B., & Osthus, D. (2025). Monotonic warpings for additive and deep Gaussian processes. <i>Statistics & Computing</i> , 35, 65. https://doi.org/10.1007/s11222-025-10598-7 McAvoy, T. J., Foley, J., Barnett, S. D. , Mays, R., Dechaine, A., & Salom, S. M. (2024). Laboratory Assessment of Predation and Fecundity of <i>Laricobius</i> Species (Coleoptera: Derodontidae), Predators of Hemlock Woolly Adelgid. <i>Biocontrol Science & Technology</i> , 34(12), 1091-1114. https://doi.org/10.1080/09583157.2024.2405155 Bukvic, A. & Barnett, S. (2023). Drivers of flood-induced relocation among coastal urban residents: Insight from the US east coast. <i>Journal of Environmental Management</i> , 325(A), 116429. https://doi.org/10.1016/j.jenvman.2022.116429	

**PAPERS IN
REVIEW**

Barnett, S. D., Gramacy, R. B., Beesley, L. J. , Osthus, D., Huang, Y., Guo, F., & Reisenfeld, D. B. (2026). Bayesian Statistical Inversion for High-Dimensional Computer Model Output and Spatially Distributed Counts. <https://arxiv.org/abs/2512.01927>

**TALKS &
POSTERS**

Key: **S** $\equiv$ Seminar; **IT** $\equiv$ Invited Talk; **CT** $\equiv$ Contributed Talk; **P** $\equiv$ Poster

Computer model calibration for large-scale spatially-distributed counts

IT	Oct 2024	Advances in Interdisciplinary Statistics and Combinatorics , Greensboro, NC
CT	Aug 2024	Joint Statistical Meetings , Portland, OR
CT	June 2024	Statistics in Quality, Industry, and Technology , Waterloo, CA
P	Feb 2024	SIAM Conference on Uncertainty Quantification , Trieste, IT
S	Aug 2023	Statistical Sciences Group, Los Alamos National Lab , Los Alamos, NM

Deep Gaussian process modeling of Poisson data

IT	Jun 2026	Quality & Productivity Research Conference , Santa Fe, NM
CT	Mar 2026	SIAM Conference on Uncertainty Quantification , Minneapolis, MN
S	May 2025	Statistical Sciences Group, Los Alamos National Lab , Los Alamos, NM
S	Nov 2024	Department of Statistics, Brigham Young University , Provo, UT
CT	Oct 2024	Fall Technical Conference , Nashville, TN
P	May 2024	Design and Analysis of Experiments Conference , Blacksburg, VA
P	Oct 2023	Virginia Tech Corporate Partners , Blacksburg, VA
P	May 2023	ASA/IMS Spring Research Conference , Banff, CA

Monotonic warpings for additive and deep Gaussian processes

CT	Oct 2025	Fall Technical Conference , Houston, TX
CT	Aug 2025	Joint Statistical Meetings , Nashville, TN
CT	Jun 2025	ASA/IMS Spring Research Conference , New York, NY

Statistical collaborations & consulting

P	Aug 2023	Joint Statistical Meetings , Toronto, CA
P	Feb 2023	Conference on Statistical Practice , San Francisco, CA
P	Oct 2022	Virginia Tech Corporate Partners , Blacksburg, VA

HONORS & AWARDS

Virginia Tech Rose Costain Award for Outstanding Citizenship (2025)
 Virginia Tech Jesse C. Arnold Award for Outstanding Teaching (2025)
 Student Travel Award, Fall Technical Conference (2025)
 Student Travel Scholarship, AMS/IMS Spring Research Conference (2025)
 Virginia Tech Gibbons Star Award for Combined Outstanding Research and Service (2024)
 Student Travel Award, Fall Technical Conference (2024)
 Student Travel Award, Joint Research Conference on Statistics in Quality, Industry, and Technology (2024)
 Virginia Tech Gibbons Star Award for Combined Outstanding Research and Service (2023)
 Virginia Tech Klaus Hinkelmann Award for Outstanding Departmental Service (2023)
 Student Travel Scholarship, AMS/IMS Spring Research Conference (2023)
 Virginia Tech John Bartko Ph.D. '62 Prize in Statistics for Outstanding Statistical Collaboration and Communication (2022)

TEACHING EXPERIENCE**Instructor of Record**

- STAT 4714 - Statistics & Probability for Electrical Engineers, VIRGINIA TECH (Fall 2024)

Teaching Assistant

- STAT 2004 - Introductory Statistics, VIRGINIA TECH (Fall 2022)
- STAT 5204G - Experimental Design: Concepts and Applications, VIRGINIA TECH (Fall 2021)
- STAT 3654 - Intro to Data Analytics and Visualization, VIRGINIA TECH (Spring 2021)
- STAT 4664 - Computational Intensive Stochastic Modeling, VIRGINIA TECH (Spring 2021)
- STAT 4654 - Intermediate Data Analytics and Machine Learning, VIRGINIA TECH (Fall 2020)
- CS 465 - Computer Security, BRIGHAM YOUNG UNIVERSITY (Winter 2016)
- CS 312 - Algorithm Design & Analysis, BRIGHAM YOUNG UNIVERSITY (Fall 2015)

OTHER RESEARCH EXPERIENCE

LOS ALAMOS NATIONAL LAB STATISTICAL SCIENCES GROUP: Graduate research intern. Developed novel Gaussian process model calibration and surrogate modeling methods for non-continuous, large-scale satellite and computer model data; 2023-2025

VIRGINIA TECH STATISTICAL APPLICATIONS AND INNOVATIONS GROUP: Lead collaborator. Provided statistical consulting and education to graduate students, faculty, and working professionals; 2021-2023

NCAR EARLY CAREER FACULTY INNOVATOR PROGRAM: Graduate research assistant. Conducted survey data analysis to understand attitudes on permanent relocation for coastal residents; Summer 2021-2022

SERVICE

ASQ, Chemical and Process Industries Division, Chair-Elect; 2026
ASQ, Chemical and Process Industries Division, Webinar/Communications Chair; 2023-2025
Mu Sigma Rho, President of Virginia Tech Chapter; 2023-2025
Mu Sigma Rho, Treasurer of Virginia Tech Chapter; 2022-2023
Virginia Tech Data Science Camp, Organizer and Facilitator; Summer 2022
Virginia Tech Corporate Partners Committee; 2021-2025

**PROFESSIONAL
MEMBERSHIP**

American Society for Quality (ASQ), Chemical and Process Industries Division
American Statistical Association (ASA), Section on Physical and Engineering Sciences
International Society for Bayesian Analysis (ISBA)
Society for Industrial and Applied Mathematics (SIAM)
Mu Sigma Rho National Honor Society